

14. Contingency Tables & Goodness-of-Fit

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14.1 Tests of Independence

Often one has a sample of cases; each case can be categorized according to two different criteria:

- each person got the drug or got a placebo, and each person lived or died;
- a criminal got the death penalty or not, and the state (AL, AZ, ...) in which they were charged;
- letter grade in a statistics course, and major

A **contingency table** shows counts for two categorical variables. For example:

	Math	English	History
Male	10	20	15
Female	20	10	15

Suppose one took the 50 U.S. states and classified them as to whether they support Romney or Obama, and how many executions they had in the last five years (e.g., 0, 1-5, more than 5). You might get a **contingency table** that looks like this:

	Obama	Romney	
0	20	1	21
1-5	5	8	13
> 5	2	14	16
	27	23	50

Here there are 20 states that supported Obama and had no executions, 1 state that supported Romney and had no executions, and so forth.

The general null and alternative hypotheses are:

H_0 : The two criteria are independent.

H_A : Some dependence exists.

For a given situation, it is always better to be clear and specific to the context of the problem. For this example, the hypotheses are:

H_0 : Voting preference has nothing to do with execution rates.

H_A : There is a relationship between voting choice and executions.

Unlike previous cases, there is only one choice for the null and alternative hypothesis. But as with all of our hypothesis tests, there are three parts. We now we need to get a test statistic and a critical value.

The test statistic is

$$ts = \sum_{\text{all cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

The O_{ij} is the observed count for the cell in row i , column j .

The E_{ij} uses the following formula:

$$E_{ij} = \frac{(\textit{ith row sum}) * (\textit{jth column sum})}{\textit{total}}$$

This is why in the example contingency table we showed the row sums, column sums, and the total count—to make the calculation of E_{ij} more easy.

For our example, we find:

$$E_{11} = 21 * 27/50 = 11.34$$

$$E_{12} = 21 * 23/50 = 9.66$$

$$E_{21} = 27 * 13/50 = 7.02$$

$$E_{22} = 23 * 13/50 = 5.98$$

$$E_{31} = 27 * 16/50 = 8.64$$

$$E_{32} = 23 * 16/50 = 7.36$$

Then the test statistic is:

$$\begin{aligned} ts &= \frac{(20 - 11.34)^2}{11.34} + \frac{(1 - 9.66)^2}{9.66} + \dots + \frac{(14 - 7.36)^2}{7.36} \\ &= 26.734. \end{aligned}$$

We compare the test statistic to the value from a chi-squared distribution with degrees of freedom equal to

$$k = (\text{number of rows} - 1) * (\text{number of columns} - 1).$$

For our example, $k = (3 - 1) * (2 - 1) = 2$.

The significance probability is

$$P\text{-value} = \mathbf{P}[W > ts]$$

where W has the chi-squared distribution with k degrees of freedom.

For a chi-squared random variable with 2 degrees of freedom, the 1% value is 9.21. So

$$.01 = \mathbf{P}[W > 9.21] > \mathbf{P}[W > 26.73] = P\text{-value}.$$

So the significance probability is much less than 0.01. At the 0.01 level, we reject the null hypothesis. There is strong evidence that political preference and execution rates are somehow connected.

But the connection can be very subtle. We cannot infer causation, and the apparent relationship may not be at all what we expect. For example, one might argue that voting preferences reflect economic hardship, and states with economic hardship experience more violent crime and thus use the death penalty more often.

Sometimes there are hidden confounders that are more interesting than the relationship between the two classification criteria. It can even happen that the hidden confounder can reverse the apparent relationship in the data. When this happens, it is called **Simpson's Paradox**.

For example, we could have made a contingency table of the criteria accept/reject versus major in the Berkeley graduate admissions data.

19.2 Goodness-of-Fit Tests

Goodness-of-fit tests are used to decide whether data accord well with a particular theory. For example, recall that Gregor Mendel was an Augustinian monk in charge of the monastery's truck garden. He noted that several traits in pea plants, e.g.:

- color
- height
- wrinkled pods

seemed to be inherited from the parent plants in a predictable way.

To study color Mendel got inbred strains, whose progeny were always yellow or always green. Then he did experiments in which those inbred strains were crossed, and he observed the colors of the offspring.

Recall from biology: Mendelian theory says that each plant has two genes for color, and each parent contributes one of those genes, at random, to the progeny. Thus:

$$GG \times GG \Rightarrow GG$$

$$YY \times YY \Rightarrow YY$$

$$YY \times GG \Rightarrow YG$$

$$YY \times YG \Rightarrow YY, YG$$

$$GG \times YG \Rightarrow GG, GY$$

$$YG \times YG \Rightarrow GG, YY, YG, GY$$



Yellow is dominant. Plant that have a yellow gene provide only yellow peas.

The inbred plants were **GG** or **YY**. When crossing these, the first generation all had yellow peas (because of dominance) even though genetic composition of each plant was **YG**.

The second generation was formed by crossing the first generation plants:

$$\text{YG} \times \text{YG} \Rightarrow \text{GG}, \text{YY}, \text{YG}, \text{GY}$$

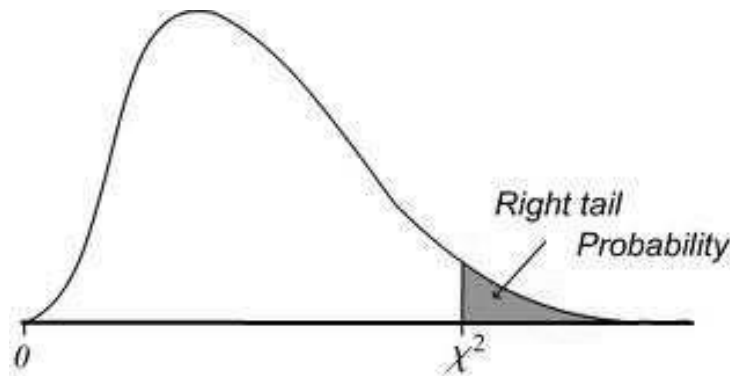
and it gave plants such that 3/4 had yellow peas, 1/4 had green peas.

Seed		Flower Color	Pod		Stem	
Form	Cotyledons		Form	Color	Place	Size
 Grey & Round	 Yellow	 White	 Full	 Yellow	 Axial pods, Flowers along	 Long (6-7ft)
 White & Wrinkled	 Green	 Violet	 Constricted	 Green	 Terminal pods, Flowers top	 Short $\frac{1}{2}$ -1ft)
1	2	3	4	5	6	7

Mendel could predict that among, say, 100 second generation offspring, about 25 should bear green peas. He made many such crosses; his predicted numbers were close to those observed. But how can Mendel prove his theory?

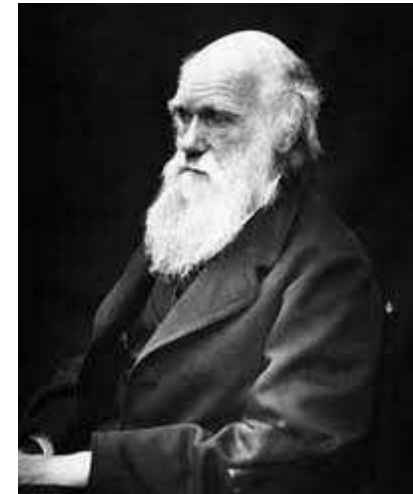
He had no statistical way to show that his observed counts of yellow and green pea plants matched well to the predictions from his model. All he could do was present his predictions, his counts, and wave his hands.

So he (probably) faked his data in order to get better agreement and thus to present a stronger case. His reported counts were too good to be true—they were closer to his predictions than could happen under his model.



Since Mendel's basic experiment had only two categories, he could have used a test of whether the proportion of green peas was $1/4$ (Chinese menu, IIc) to assess his theory. But we want to handle cases that are more complicated, so consider an inheritance experiment that Darwin performed.

Darwin studied peonies, in which color inheritance is co-dominant or additive. Specifically, he crossed red and white peonies and got all pink. Then he crossed pinks with pinks and got some red, some white, and some pink.



Mendel and Darwin needed a way to assess the statistical significance of such predictions. Are the observed numbers too far from the numbers predicted by Mendel's theory? Or are the numbers close enough to agree with Mendel's model for inheritance?

For this example, we want to know whether the counts of red, white and pink peonies agree closely with the $1/4: 1/4: 1/2$ ratios that are predicted.

In this type of test, the null and alternative are always the same. They are:

H_0 : The model holds vs. H_A : The model fails.

In particular applications one can be more specific; e.g.:

H_0 : The ratios of red, white and pink are $1/4: 1/4/: 1/2$

H_A : The ratios differ from $1/4: 1/4/: 1/2$.

Note that we can only reject the model. We cannot prove it, since we never “prove” the null hypothesis. The best we can do is to fail to reject it.

Our test statistic is similar to that for contingency tables (because testing for independence is testing for a specific kind of model). Here, the test statistic is:

$$ts = \sum \frac{(O_i - E_i)^2}{E_i}$$

where the sum is taken over all categories (i.e., red, white, and pink). The O_i is the observed count in category i , and E_i is the count predicted in that category by the model.

To be concrete, suppose Darwin had made 100 crosses of pink with pink and had gotten 22 red, 29 white, and 49 pink. So $O_1 = 22$, $O_2 = 29$, and $O_3 = 49$.

The expected counts are those predicted by the model. Thus $E_1 = 25$, $E_2 = 25$, and $E_3 = 50$.

The numerical value of the test statistic is

$$\begin{aligned} ts &= \frac{(22 - 25)^2}{25} + \frac{(29 - 25)^2}{25} + \frac{(49 - 50)^2}{50} \\ &= 1.02. \end{aligned}$$

The significance probability comes from a chi-squared table. Let W be a chi-squared random variable with

$$k = \text{\#categories} - 1$$

degrees of freedom. In this example, $k = 3 - 1 = 2$.

The significance probability is:

$$P - \text{value} = \mathbf{P}[W \geq ts] = \mathbf{P}[W \geq 1.02].$$

From the table, this is between .7 and .5. So the null is not rejected. The data support Mendel.